

PRACTICAL EXPERIMENT – 1

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Section: 08

Skill: Process Creation, System Calls, Parent–Child Process Management

Title

Implementation of Process Creation and Command Execution in Linux using fork(), exec(), and wait() System Calls.

Aim

To develop a C program that accepts a Linux command from the user, creates a child process using fork(), executes the command, waits for the child process, and displays Parent and Child PIDs.

Theory

Process creation in Linux can be performed using the fork() system call. The fork() system call creates a new child process. The child process can execute a Linux command using an exec() family system call such as execlp(). The parent process can wait for the child process to complete using wait(). Process IDs (PIDs) are obtained using getpid() and the parent PID using getppid().

Algorithm

1. Start the program.
2. Accept a Linux command from the user.
3. Remove the newline character from the input.
4. Create a child process using fork().
5. If fork() fails, display an error message and terminate.
6. In the child process, display the Child PID and Parent PID.
7. Execute the entered Linux command using execlp().
8. In the parent process, display the Parent PID.
9. Wait for the child process to complete using wait().
10. Display the child process completion message.
11. Stop the program.

Code

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    printf("OSSP Experiment 1\n");
    printf("Shell Architecture\n");
    printf("Parent PID: %d\n", getpid());

    pid_t pid = fork();
```

```

if (pid == 0)
{
    printf("Child Process PID: %d\n", getpid());
}
else if (pid > 0)
{
    wait(NULL);
    printf("Parent Process: Child completed\n");
}
else
{
    printf("Fork failed\n");
}

return 0;
}

```

Execution Screenshots

Main C Code Creation

```

admin@heeraj@DESKTOP-J39GHP7:~$ cat > main.c <<'EOF'
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    printf("OSSP Experiment 1\n");
    printf("Shell Architecture\n");
    printf("Parent PID: %d\n", getpid());

    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child Process PID: %d\n", getpid());
    }
    else if (pid > 0)
    {
        wait(NULL);
        printf("Parent Process: Child completed\n");
    }
    else
    {
        printf("Fork failed\n");
    }

    return 0;
}
EOF

```

Program Execution using lscpu

```

admin@heeraaj@DESKTOP-J39GHP7:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 8
On-line CPU(s) list:   0-7
Vendor ID:              GenuineIntel
Model name:             Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz
CPU family:             6
Model:                 165
Thread(s) per core:     2
Core(s) per socket:     4
Socket(s):              1
Stepping:               2
BogoMIPS:               4992.01
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss
                        e sse2 ss ht syscall nx pdpe1gb rdtscp lm arch_perfmon rep_good nopl xtopology cpuid tsc_kn
                        own_freq pni pclmuldq vmx ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 x2apic movbe popcnt aes x
                        save avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhance
                        d tpr_shadow ept vpid ept_ad fsgsbase bti avx2 smep bmi2 erms invpcid rdseed adx smap clfl
                        ushopt xsaveopt xsavec xgetbv1 xsaves vnm1 md_clear flush_lld arch_capabilities

Virtualization features:
  Virtualization:       VT-x
  Hypervisor vendor:    Microsoft
  Virtualization type:   full

Caches (sum of all):
  L1d:                  128 KiB (4 instances)
  L1i:                  128 KiB (4 instances)
  L2:                   1 MiB (4 instances)
  L3:                   8 MiB (1 instance)

NUMA:
  NUMA node(s):         1
  NUMA node0 CPU(s):    0-7

Vulnerabilities:
  Gather data sampling:  Unknown: Dependent on hypervisor status
  Ghostwrite:           Not affected
  Indirect target selection: Mitigation; Aligned branch/return thunks
  Itlb multihit:        KVM: Mitigation: Split huge pages
  L1tf:                 Not affected
  Mds:                  Not affected
  Meltdown:             Not affected
  Mmio stale data:       Vulnerable: Clear CPU buffers attempted, no microcode; SMT Host state unknown
  Old microcode:         Not affected
  Reg file data sampling: Not affected
  Retbleed:             Mitigation; Enhanced IBRS
  Spec rstack overflow:  Not affected
  Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:            Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-IBRS SW sequence; BHI SW lo
                        op, KVM SW loop
  Srbds:                Unknown: Dependent on hypervisor status
  Tsa:                  Not affected
  Tsx async abort:       Not affected
  Vmscape:              Not affected

```

Program Execution using uname

```

admin@heeraaj@DESKTOP-J39GHP7:~$ uname
Linux

```

Program Execution using lsblk

```

admin@heeraaj@DESKTOP-J39GHP7:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0     2G   0 disk [SWAP]
sdd   8:48   0     1T   0 disk /mnt/wslg/distro

```

Program Execution using ps

```
admin@heeraj@DESKTOP-J39GHP7:~$ ps
PID TTY          TIME CMD
561 pts/2        00:00:00 bash
883 pts/2        00:00:00 ps
```

Program Execution using top

```
883 pts/2        00:00:00 ps
admin@heeraj@DESKTOP-J39GHP7:~$ top
top - 04:58:43 up 18 min, 1 user, load average: 0.01, 0.04, 0.08
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7859.6 total, 7323.9 free, 511.5 used, 174.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7348.1 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24184   15408  11624  S   0.0   0.2   0:01.26 systemd
    2 root        20   0    3180    2208   2072  S   0.0   0.0   0:00.01 init-systemd(Ub
    8 root        20   0    3180    2044   1940  S   0.0   0.0   0:00.00 init
   48 root       19  -1   50400   16776   15480  S   0.0   0.2   0:00.29 systemd-journal
   83 systemd+   20   0   22416   14480   11984  S   0.0   0.2   0:00.13 systemd-resolve
   91 root       20   0   35344   12284    9216  S   0.0   0.2   0:01.11 systemd-udev
  112 root       20   0    2888    1948    1820  S   0.0   0.0   0:00.07 chronyd-starter
  113 root       20   0    4472    3116    2868  S   0.0   0.0   0:00.01 cron
  114 message+   20   0    8816    5408    4732  S   0.0   0.1   0:00.08 dbus-daemon
  121 root       20   0   44096   29856   14712  S   0.0   0.4   0:00.53 networkd-dispat
  161 root       20   0   18436    9432    8184  S   0.0   0.1   0:00.16 systemd-logind
  180 syslog     20   0   220548    5720    4496  S   0.0   0.1   0:00.13 rsyslogd
  244 root       20   0    5492    2744    2564  S   0.0   0.0   0:00.02agetty
  288 _chrony     20   0   24048   11572    7224  S   0.0   0.1   0:00.12 chronyd
  293 _chrony     20   0   12096    2404    1760  S   0.0   0.0   0:00.00 chronyd
  323 root       20   0  123460   32736   18128  S   0.0   0.4   0:00.29 unattended-upgr
  378 root       20   0    8648    5512    4668  S   0.0   0.1   0:00.01 login
  425 admin@h+   20   0   22344   13604   10980  S   0.0   0.2   0:00.23 systemd
  430 admin@h+   20   0   22912    4120    2120  S   0.0   0.1   0:00.00 (sd-pam)
  459 admin@h+   20   0    6136    5408    3856  S   0.0   0.1   0:00.06 bash
  556 root       20   0    3184    1120     980  S   0.0   0.0   0:00.00 SessionLeader
  557 root       20   0    3200    1260    1108  S   0.0   0.0   0:00.03 Relay(561)
  561 admin@h+   20   0    6264    5536    3864  S   0.0   0.1   0:00.16 bash
  895 admin@h+   20   0    8164    6056   3964  R   0.0   0.1   0:00.01 top
```

Result

The program was executed successfully. Linux system and process information was displayed through the required Ubuntu terminal commands, and the process creation demonstration was executed successfully.